



Software Development Engineering Training Curriculum

Module 1: Web & JavaScript Foundations

- **Hours: 15**
- **Lessons:**
 - HTML5 & CSS3 (3 hrs)
 - Responsive Layouts (Flexbox & Grid) (2 hrs)
 - Tailwind CSS (2 hrs)
 - JavaScript Fundamentals (ES6+) (5 hrs)
 - DOM Manipulation & Events (2 hrs)
 - Async JavaScript (Promises, Fetch API) (3 hrs)
 - Browser Developer Tools (2 hrs)
- **Outcomes:**
 - Build responsive static websites
 - Understand core JS concepts and DOM operations

Module 2: Git & Version Control

- **Hours: 4**
- **Lessons:**
 - Git Basics and CLI (1.5 hrs)
 - Branching, Merging, and Pull Requests (1.5 hrs)
 - Collaboration using GitHub (1 hr)
- **Outcomes:**
 - Manage codebases using Git
 - Collaborate with teams via GitHub

Module 3: React Essentials

- **Hours: 16**
- **Lessons:**
 - Introduction to React & JSX (2 hrs)
 - Functional Components, Props, and State (3 hrs)
 - Lists and Conditional Rendering (2 hrs)
 - Event Handling and Forms (3 hrs)
 - React Developer Tools (1 hr)
 - Small React Projects (5 hrs)
- **Outcomes:**
 - Build dynamic UIs with React
 - Create interactive components

Module 5: React Intermediate + Routing & Hooks

- **Hours: 14**
- **Lessons:**
 - React Router (2 hrs)
 - useEffect and useContext (3 hrs)
 - Custom Hooks (2 hrs)
 - useReducer & useRef (3 hrs)
 - Next Js (3 hrs)
 - Project: Blog or Recipe App (4 hrs)
- **Outcomes:**
 - Implement routing and stateful logic
 - Build multi-page SPAs

Module 6: React Native Fundamentals

- **Hours: 14**
- **Lessons:**
 - Setup with Expo & CLI (2 hrs)
 - React Native Components & Styling (3 hrs)
 - Navigation & Screens (2 hrs)
 - Device APIs: Camera, Location (2 hrs)
 - Mini Project: Mobile Todo App (5 hrs)
- **Outcomes:**
 - Build basic mobile apps

- Use native device APIs

Module 7: Backend Development with Node.js & Express

- **Hours: 18**
- **Lessons:**
 - Node.js & Express Basics (3 hrs)
 - REST API Design (3 hrs)
 - PostgreSQL and ORM (3 hrs)
 - MongoDB with Mongoose (3 hrs)
 - Authentication with JWT (3 hrs)
 - Middleware & Error Handling (2 hrs)
 - Project: API for Todo or Blog (4 hrs)
- **Outcomes:**
 - Create REST APIs
 - Handle auth and data persistence

Module 8: TypeScript, Python & Django Essentials

- **Hours: 16**
- **Lessons:**
 - TypeScript for React Developers (3 hrs)
 - TypeScript for Nest Js (3 hrs)
 - Python Basics & OOP (4 hrs)
 - Django Fundamentals & Admin (3 hrs)
 - Django REST Framework (3 hrs)
 - Comparison: Node.js vs Django (3 hrs)
- **Outcomes:**
 - Understand statically typed JS
 - Build fullstack apps with Django

Module 9: Fullstack Project I (Web + Mobile)

- **Hours: 20**
- **Project:**
 - Build a social media app with React frontend, Node.js backend, React Native mobile version
 - User authentication, CRUD features, state management
- **Outcomes:**
 - Integrate fullstack architecture
 - Deploy frontend/backend

Module 10: Agile Software Development

- **Hours: 10**
- **Lessons:**
 - Agile Manifesto and Principles (2 hrs)
 - Scrum Framework: Roles, Events, and Artifacts (3 hrs)
 - Agile Ceremonies: Sprint Planning, Standups, Reviews (2 hrs)
 - Working in Agile Teams (2 hrs)
 - Agile Tools: Jira, Trello (1 hr)
- **Outcomes:**
 - Understand Agile methodology and Scrum framework
 - Apply Agile practices in software development teams

Module 11: Cloud Native Architecture

- **Hours: 18**
- **Lessons:**
 - 12-Factor App & Microservices (3 hrs)
 - Docker & Docker Compose (4 hrs)
 - Kubernetes Fundamentals (5 hrs)
 - Service Discovery, Config Maps, Secrets (3 hrs)
 - Helm & K8s Deployment (3 hrs)
 - Serverless Functions and Netlify
- **Outcomes:**
 - Build scalable, containerized apps
 - Deploy microservices with Kubernetes

Module 12: CI/CD, CDN & Caching

- **Hours: 12**
- **Lessons:**
 - GitHub Actions Basics (3 hrs)
 - Linting, Testing, Building (2 hrs)
 - Deployment to AWS/GCP (2 hrs)
 - Using CDNs for static assets (2 hrs)
 - Introduction to caching (Redis, HTTP caching, CDN strategies) (3 hrs)
- **Outcomes:**
 - Automate deployments
 - Improve performance with CDNs and caching techniques

Module 13: System Design, Patterns & Testing

- **Hours: 20**
- **Lessons:**
 - Clean Code Principles (2 hrs)
 - Design Patterns in JS (3 hrs)
 - SOLID, KISS, DRY, YAGNI (3 hrs)
 - Unit & Integration Testing (Jest, Cypress) (5 hrs)
 - System Design Fundamentals (APIs, Load Balancing) (5 hrs)
 - Project: Design Doc + Sample System (2 hrs)
- **Outcomes:**
 - Apply software engineering principles
 - Design scalable systems

Module 14: Prompt Engineering for Developer Productivity

- **Hours: 6**
- **Lessons:**
 - Introduction to Prompt Engineering (1 hr)
 - Using AI tools for code generation and debugging (2 hrs)
 - Patterns: Recipe, Persona, Chain of Thought, Few-Shot (2 hrs)
 - Integrating AI into dev workflow (1 hr)

- **Outcomes:**
 - Use ChatGPT/AI tools to accelerate development
 - Apply structured prompts for high-quality coding support

Module 15: Data Structures and Algorithms

- **Hours: 15**
- **Lessons:**
 - Introduction to DSA and Big-O Notation (1 hr)
 - Arrays, Strings & Hashmaps (2 hrs)
 - Linked Lists & Stacks/Queues (2 hrs)
 - Recursion & Tree Traversal (2 hrs)
 - Sorting & Searching Algorithms (2 hrs)
 - Graphs & Pathfinding (BFS, DFS, Dijkstra's) (2 hrs)
 - Dynamic Programming & Memoization (2 hrs)
 - Real-world Use Cases + Mock Interview Problems (2 hrs)
- **Outcomes:**
 - Understand core data structures and algorithm types
 - Analyze and optimize time/space complexity
 - Solve real-world and interview-style problems efficiently
 - Improve confidence in coding interviews and system design

Module 16: Capstone Project

- **Hours: 25**
- **Project:**
 - Build and deploy a SaaS app (Web + Mobile)
 - Use React, React Native, GraphQL, Nest.js or Django
 - Deploy using Docker + Kubernetes + CI/CD
- **Outcomes:**
 - Showcase in portfolio for job readiness